An Evolutionary Safety Dial for Frozen Manipulation Policies: A Physically Audited Pilot

Execution pilots, forecast audit and counterexample premise, 19 October 2026

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ABSTRACT

Can a compact execution controller improve safe completion around a frozen manipulation policy? In ManiGuard Clutter, Part 1 allocates 24 simulator episodes to evolutionary search and 24 fresh episodes to one frozen finalist at three dial strengths. Its training lead does not persist: identity safely completes 5/8 episodes with no observed violation, versus 3/8 with two violations at half strength and 4/8 with two violations at full strength. Part 2 separates query timing from restraint in sixteen new episodes. Identity, fixed-eight and correction-triggered execution each safely complete 3/4, versus 2/4 for fixed-four, with one violation per condition. The adaptive trigger never refreshes early; fixed-four more than doubles query cost without improving measured safe completion. Part 3 prospectively tests a frozen visual dynamics ensemble before allowing a matched evolutionary allocation; its forecast audit fails accuracy despite adequate event support, gating the physical optimizer comparison. Four-action error is 45.1% worse than persistence, and correction calibration collapses to ordinary disagreement trust. Part 4 then tests a direct uprightness-event predictor on 24 verified historical trajectories. Its development operating point alarms on 2,045/2,229 negative windows, with no confident miss among one first-event episode; its frozen gate stops physical counterexample acquisition before new simulator interaction. We preserve causal executed-action histories, native outcomes, videos and continuous allocation accounting. These descriptive pilots establish neither optimizer superiority nor calibrated physical risk.

CCS CONCEPTS

• **Computing methodologies**;

KEYWORDS

evolutionary search, execution adaptation, manipulation, safety, frozen policies

1 QUESTION AND SCOPE

Task completion can conceal a manipulation policy's physical specification violations. ManiGuard separates task success from safety using runtime monitors over physics-grounded predicates [7]. We test whether a small execution controller can improve safe completion around a frozen Clutter policy, and whether an external dial exposes a useful measured trade-off. All physical outcomes here are measured in simulation. The monitor labels actions' consequences; it does not filter commands.

Part 1 implements its plan at planning commit `8df7ac9`. The machine-readable protocol was frozen before search. It contains the controller domain, information restrictions, source hashes, scene partitions, schedules, selection rules and absolute deadlines. The work establishes an executable and auditable experiment. It provides no optimizer-superiority or safety-guarantee claim.

2 PART 1: CONTROLLER AND FROZEN PROTOCOL

The genome has four categorical coordinates: free-motion joint-target cap $\{\varnothing, .12, .08, .04\}$ rad, grasp cap $\{\varnothing, .08, .04, .02\}$ rad, smoothing $\{1, .75, .5\}$, and query horizon $\{4, 8, 16\}$. This defines 144 points. The neutral genome disables caps and uses smoothing one and horizon eight.

For proposed arm targets p, measured joints q and smoothing α, define $r = q$ on the first action and the previous actual environment input thereafter. The anchor computes

$$z = r + \alpha(p - r), \qquad a = q + \mathrm{clip}(z - q, -c, c), \tag{1}$$

with $a = z$ when the applicable cap is disabled. The fixed causal finger heuristic selects the grasp cap: absolute finger positions are in $[.003, .035]$, their maximum change is below .002, and the proposal closes the gripper. It does not identify the held object.

The episode-constant external dial d applies $p + d(a - p)$ to the arm and uses

$$h(d) = \lfloor 8 + d(h_\theta - 8) + .5 \rfloor . \tag{2}$$

Zero bypasses arithmetic exactly; one returns the anchor directly. Native gripper binarization, downstream clipping and termination remain intact. History updates from the final clipped environment input, before low-level actuator processing. The controller uses only proposals, measured arm/finger positions and that causal history. Object states, goal labels and monitor predicates are excluded. Caps bound target displacement rather than force or collision probability.

Search runs only at $d = 1$, using PCG64 seed 20261018. Population zero contains identity and three distinct uniform random points. Each of three populations has four slots, evaluated on two training assignments. The next population copies the best point for fresh physical re-evaluation and creates three mutated children from the best two parents. The declared categorical mutation kernel rejects unchanged children and within-population duplicates; stable birth order resolves ties, with the copied elite first. Fitness

ManiGuard pilot, 19 October 2026,
2026.

Table 1: Every population's mean candidate utilities, in birth order. The elite is physically re-evaluated rather than retaining historical fitness.

Population	Utilities	Safe complete	Violating
0	.5, 1, .5, 1	6/8	0/8
1	1, 1, 0, .5	5/8	0/8
2	.5, 1, .5, −.5	5/8	1/8

is

$$\overline{U} = \mathrm{mean}\big[\mathbf{1}(S_{\mathrm{req}}) - 2\mathbf{1}(V)\big], \tag{3}$$

where V is raw `ltl_violated`, C_{req} combines native success with the pinned Clutter checklist, and $S_{\mathrm{req}} = C_{\mathrm{req}} \wedge \neg V$. Infrastructure failures have no fitness. Only the final evaluated population can supply the finalist.

Engineering uses `task_0040/base`; search uses tasks 0034 and 0002; comparison uses tasks 0029 and 0032. Whole-task partitions are preserved and test scenes remain untouched. Training base seeds are $1000 + 10g + i$; comparison uses $2000 + 10j + r$ with four repetitions per scene. The native evaluator derives sampling seeds as CRC32 of the base seed and scene, masked to 31 bits. The server resets a stream for each episode ID. Independent schedule seed 20261019 seals adjacent three-dial blocks, cycling all six permutations before repetition. Scenes are conservatively treated as exposed; the comparison tests fresh repetitions.

3 PART 1: SEARCH AND FRESH COMPARISON

Population-zero timing selected the full 48-episode allocation before selection inspected scores. The shared eight episodes count once. The planning duration was $\max(417\ \mathrm{s}, 1.25 \times \mathrm{slowest}) = 417$ s; the remaining estimate was 16,980 s against 21,149.97 s available. The predefined 28-episode fallback was unused. All full jobs retain the native 1,100-action horizon and native early success.

The selected genome is $[3, 3, 0, 2]$: free cap .04 rad, grasp cap .02 rad, smoothing one, horizon sixteen. It is a new offspring of population one's second eligible parent. It safely completes both final training episodes. The copied elite scores only .5 in its last re-evaluation (Table 1, Figure 2). One discarded search candidate completes unsafely; its outcome and costs remain. The finalist's file hash is sealed before comparison, and all three settings use it.

Fresh safe completion is 5/8 at identity, 3/8 at midpoint and 4/8 at full strength (Figure 1). Full minus identity is $-1/8$, or -12.5 percentage points, with two additional violating episodes. Full strength achieves six total requested completions, but two are unsafe. Benchmark success and requested completion agree on all 24 comparison episodes, with separate labels preserved. No setting is selected or tuned against these outcomes.

On task 0029, safe completions are 1/4, 0/4 and 1/4. On task 0032, they are 4/4, 3/4 and 3/4. Each adapted setting records one violation on each scene. The first failing predicate is `container_upright=false` on task 0029 and `obstacles_upright=false` on task 0032. Both full-strength violating episodes later reach the goal. The three-point response

Table 2: Fresh interaction and process costs. Changed commands compare exact numerical inputs; cap activation is reported separately in the audit.

Dial	Actions	Queries	Process (s)	Changed
0	6,160	774	1,014.0	0.00%
.5	6,656	558	1,043.6	4.51%
1	4,928	312	873.5	8.36%

Table 3: Complete attempt accounting. Physics covers post-application callbacks; pre-subscription advances are unknown for all 52 attempts.

Work	Attempts	Actions	Queries	Physics ticks
Engineering	4	96	19	920
Search	24	17,318	1,738	105,972
Comparison	24	17,744	1,644	108,528
Total	52	35,158	3,401	215,420

supports no calibrated or monotonic physical-risk interpretation (Figure 3).

Query latency totals are 76.8, 55.2 and 31.3 s respectively. Lower query cost at full strength accompanies worse measured safety and safe completion; changing commands and query horizon together prevents a mechanism-specific causal claim. Contact/grasp episode counts, each out of eight, are $(5, 5)$, $(5, 4)$ and $(7, 6)$. The grasp heuristic activates on 955 midpoint and 1,325 full-strength actions; anchor caps activate on 235 and 350. Identity's diagnostic anchor counters do not change commands. Non-engagement in several unfinished episodes limits interpretation of zero violations alone.

4 PART 1: ENGINEERING AND PROVENANCE

CPU verification passes 33 Part 1 tests and 109 regression tests, with four skips. Checks cover identity, causal clipped-action history, caps, finger switching, interpolation, query schedules, seeded evolution, evidence corruption and owned process-group cleanup. Three successful engineering checks use 32 actions each and 4/6/8 queries for an engineering horizon-four anchor. Chunks support horizon sixteen. One earlier check fails after one completed query and zero policy actions because strict JSON rejects infinite native action bounds. Logging is fixed before search, without changing clipping. A naming bug produces nested engineering paths; these are preserved and the coordinator is repaired before search. All 48 full episodes pass physical trace replay and provenance audits.

All 3,401 queries complete; the failed check is a logging failure. There are 35,678 environment-step attempts and returns, 1,795.17 recorded simulated seconds, 360.03 s of query latency and 5,982.17 s of outer episode-process time. Counts include all initialization, discarded candidates and copied elites. Pre-subscription physics is unknown, not zero; failed-attempt counters remain lower bounds. Overlapping server durations do not add allocated GPU-hours.

The user-confirmed replacement is Vast instance 56582094 with one RTX 4090 and driver 575.57.08, verified through the kernel,

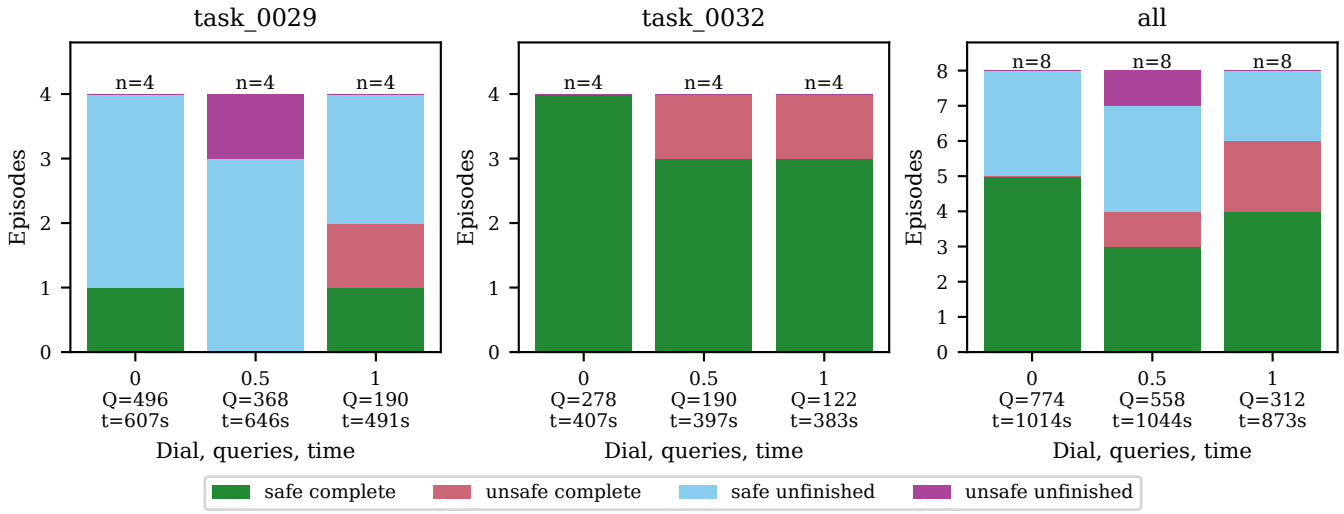


Figure 1: Every fresh outcome, per scene and pooled. Each scene has four episodes per setting; pooled denominators are eight. Q is policy queries and t is summed outer episode-process time. Unsafe completion remains separate from safe completion.

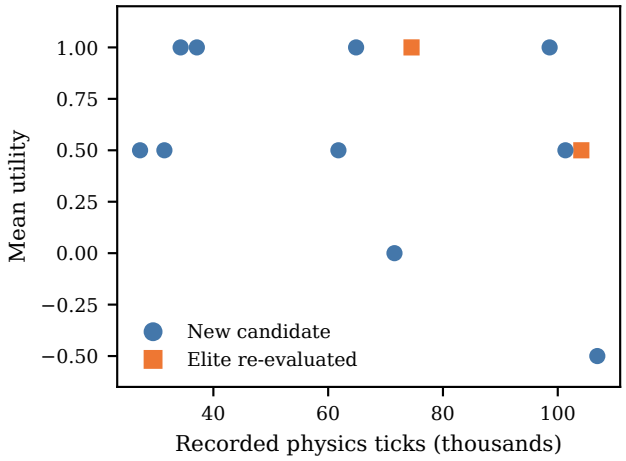


Figure 2: All twelve candidate measurements versus cumulative recorded physics ticks, including engineering and the failed check. Squares distinguish re-evaluated elites. These mixed-candidate measurements are not a single policy's estimated rate.

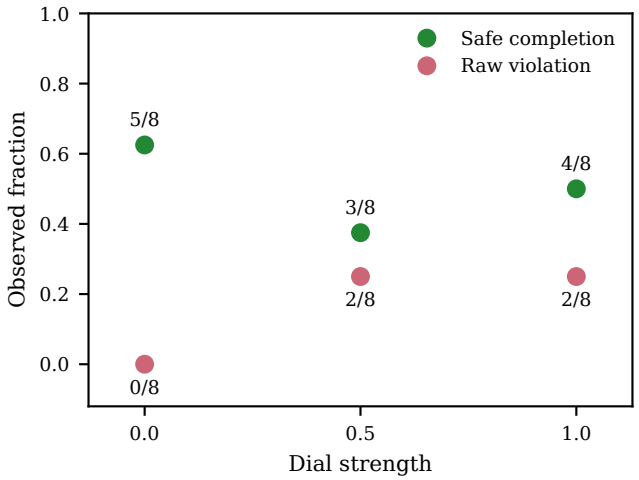


Figure 3: Measured safe-completion and violation fractions with explicit denominators. Points are descriptive; no continuous risk curve is fitted.

`nvidia-smi` and provider. The originally requested 56578764 identity is retained as provenance. No host driver is changed. Five-second memory samples reach 17,714 MiB (17.3 GiB), without bounding unobserved peaks. CPU/memory quotas are 30.6 cores and approximately 299 GiB.

Separate uv-managed Python 3.10 simulation and Python 3.11 policy environments use the pinned ManiGuard, BEHAVIOR and openpi stack [8]. The checked legacy renderer supplies 256×256 RGB overview/wrist inputs with action/render/physics rates 20/20/120 Hz, raw joint control and assisted grasping. The root environment and UNISafe installers are not used. Clutter checkpoint revision `f572e33bdb49` (full revision in the controller export), step 7100, keeps native inference settings. Controller exports retain all eighteen checkpoint-file hashes and code pins.

The original allocation clock starts at 16:00:37 UTC, including setup, implementation, failed and idle work before and after replacement. During Part 1, an independent supervised guard preserves the original 23:00:37 UTC physical target and 00:00:37 UTC next-day hard end across restarts. The policy stops at 18:25:29 UTC; process/GPU checks confirm complete exit by 18:25:57 UTC, at 2.422 allocated GPU-hours. Subsequent paper preparation, backup and idle rental continue to count. The public audit records the later accounting cutoff and rental-stop scope.

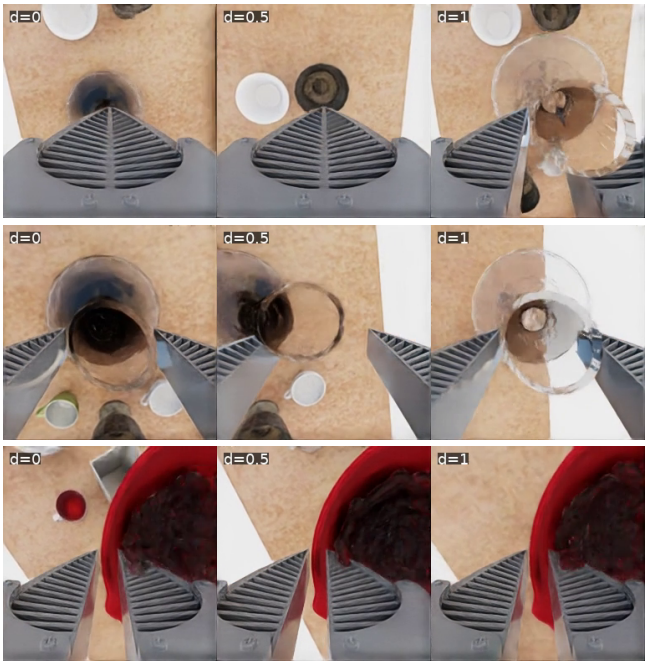


Figure 4: Actual wrist-video frames at 18 s; columns are dial 0, .5 and 1. Top: seed 2000, where full strength later safely completes while the other settings time out. Middle: seed 2002, where midpoint and full strength lose container uprightness; full strength later completes unsafely. Bottom: seed 2013, where full strength loses an obstacle's uprightness before its unsafe completion. These illustrations were selected after outcomes. All sixteen fixed-time camera strips and all sixteen complete three-setting videos are saved, covering every block. No physics is replayed.

The evidence contains 104 native video files, including the failed check, and sixteen matched composites covering both cameras in all eight comparison blocks. CPU composition preserves frame order and marks ended episodes with black frames. A 52-row public ledger, lineage, fixed-time picture gallery, frozen protocol and controller, and all figure data accompany this note. Credential-scanned raw evidence is preserved in the approved private dataset, with asset bundles, checkpoints and keys excluded. The matched-video gallery requires the owner's access. Git contains derived reporting artifacts rather than raw run directories.

5 PART 1: INTERPRETATION

The training lead disappears in the fresh check. The controller increases observed violations and does not improve safe completion. It is a working interface with an adverse pilot result. Matching seeds does not equate observations: all sixteen initial state comparisons are numerically identical, but every camera-image comparison differs. Two exposed scenes, eight repetitions per setting and one optimizer seed support only a descriptive conclusion. Identity's zero violations would give the approximate 95% upper bound $3/8 = 37.5\%$ even under independent homogeneous sampling; repeated scenes further limit interpretation.

These findings motivate isolating query scheduling from caps before increasing evolutionary search. Ordinary policy learning remains necessary for the broader research claim. Part 1 contains no such comparison or outcome-dependent extra episode. Parts 2 and 3 follow separately frozen plans at planning baseline `d4b8182`.

6 PART 2: ISOLATE CORRECTION AND FEEDBACK

Part 1 changes restraint and query horizon together. Its failure investigation found some corrected commands followed by cached proposals, motivating a bounded mechanism test rather than an established diagnosis. Four conditions use the same frozen VLA: A preserves its commands and queries every eight actions; B applies the finalist's .04/.02 rad caps at interval eight; C uses those caps at interval four; D uses those caps with correction-triggered refresh between four and eight. The finger heuristic, componentwise mapping and smoothing one are common to B/C/D. There is no new evolutionary search or three-setting dial sweep in Part 2.

After actual environment input u_t, D accumulates

$$D_k = \sum_{t \text{ since query}} \frac{\|p_t^{\text{arm}} - u_t^{\text{arm}}\|_2}{\sqrt{7}\,(.04 \text{ rad})}. \tag{4}$$

Per-coordinate differences no larger than 10^{-6} rad are zeroed first. The accumulator resets on a query, preserving previous actual input and finger history. Goal termination takes precedence; otherwise refresh occurs once $k \geq 4$ and $D_k \geq .5$, or at $k = 8$. Both the correction and trigger use already executed inputs after downstream action-space clipping. Command departure measures neither actual displacement nor physical risk. No privileged held label, object pose or monitor verdict enters this controller.

The frozen allocation has sixteen full episodes: tasks 0039/0010 each receive two fresh base seeds (4000/4001 and 4010/4011), with four conditions in each matched block. The prospective order is ABCD, BCDA, CDAB, DABC. Native horizon, goals, monitor and policy inference remain unchanged. Engineering checks A/C/D at 32 actions each; one pre-application import failure and two coordinator/tooling failures are retained. There are no full-episode retries or outcome-dependent additions.

6.1 Part 2 measured outcomes

All sixteen full episodes pass independent action/monitor replay, source/checkpoint and native sampling-seed checks. Safe completion is 3/4 for A, B and D and 2/4 for C; each condition has one violating episode (Figure 5). Requested completion is 3/4 in every condition. A/B/D each have one unsafe unfinished episode; C has one unsafe completion and one safe unfinished episode. All conditions safely complete both task 0010 episodes. On task 0039, A/B/D each safely complete one of two, whereas C safely completes neither. These small descriptive counts support no claim that one schedule is superior or that frequent feedback itself causes harm.

The adaptive rule never changes the query schedule: every one of D's 337 actual inter-query intervals is eight actions (Figure 6). Its correction threshold is crossed once, at action eight, when a query is already due. There is no crossing at actions four through

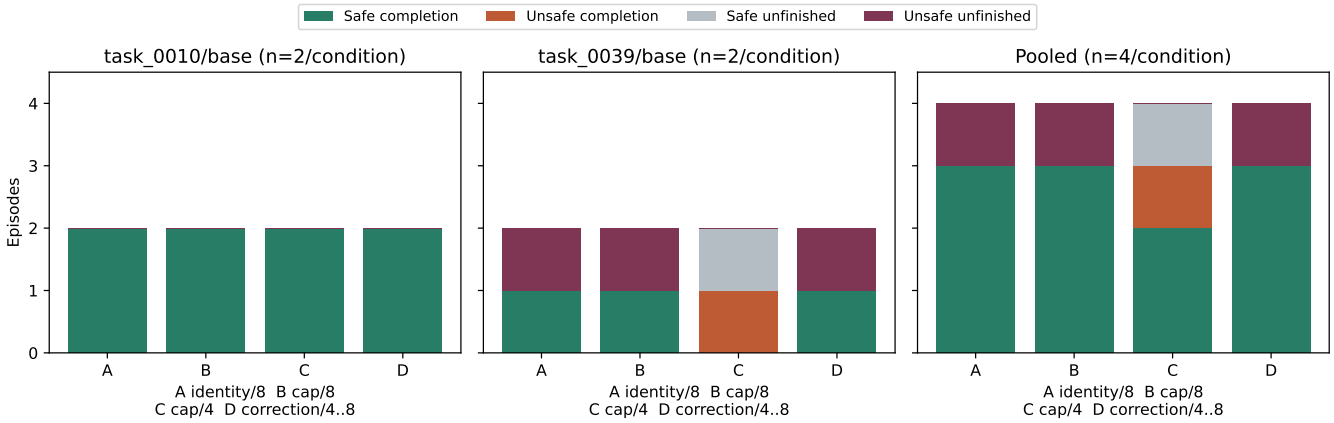


Figure 5: Part 2's complete sixteen-episode sample, with unsafe completion and both unfinished categories preserved. Each scene has two episodes per condition; pooled denominators are four. A is identity/8, B is cap/8, C is cap/4 and D is correction/4–8.

Table 4: Part 2 costs. Time is measured inside each evaluator, including capture; outer process time over all sixteen episodes is 1,897.69 s.

Condition	Actions	Queries	Time (s)	Corrections
A	2,810	353	455.13	0
B	2,674	336	438.27	76
C	2,992	750	499.49	55
D	2,714	341	451.85	54

seven. Its largest accumulated correction is .60146. Thus this run supplies no adaptive-feedback gain while leaving an actively triggering rule untested. The threshold is retained rather than adjusted against these outcomes. C uses 750 queries versus B's 336 without improving measured safe completion.

Table 4 counts substantial arm changes above 10^{-6} rad. Exact changed-command counters are 0/122/95/92; the additional numerical changes are roundoff rather than further substantial cap activations. All four episodes per condition show contact; grasp flags occur in 4/4 for A and 3/4 for B/C/D. Such flags do not establish successful continuation after contact.

All twelve initial robot-state comparisons against A are exactly equal, but none of the corresponding 24 camera pairs or twelve first chunks is exactly equal. Matched scene and sampling-seed settings therefore do not identify single-episode causal counterfactuals. The prescribed fixed-reference selection chooses B, with mean $U = .25$ versus C's zero; no additional setting is tuned.

The full sample records 11,190 controls, 1,780 queries and 68,516 post-subscription physics ticks. Engineering adds 96 controls, sixteen queries and 834 known ticks; the pre-application failure retains unknown native counters. Synthetic CPU checks precede engineering, but recorded physical-input replay first audits engineering jobs rather than preceding them as requested by the plan. This chronology deviation is explicit; all engineering replay passes before the full comparison freeze. Twenty-four targeted CPU checks pass across Parts 2 and 3. Every native camera clip, all eight matched Part 2 videos and complete event-picture strips are preserved.

7 PART 3: MODEL FEASIBILITY BEFORE EVOLUTIONARY USE

The proposed contribution concerns which motion mutants merit physical evaluation, using observed errors conditional on execution correction. Dynamics-assisted skill search already exists [4]; frozen visual-feature prediction and short rollouts also have precedent [3, 9]. These ingredients and an optimizer substitution do not establish novelty. We test a compact feasibility stage before spending the physical optimizer-comparison allocation.

Data and information. The Part 1 training traces lack later exact images and numerical margins; exposed Part 1 comparison episodes are excluded. Before fitting, the declared eight-episode recording option is sealed: B/C on tasks 0046/0006, two fresh seeds per task. All sixteen Part 2 episodes on 0039/0010 are fitting data; four 0046 episodes calibrate; four 0006 episodes audit. Whole tasks, seeds and trajectories remain separated. The 0048/0050 final comparison and main-study test partition are outside development. Acquisition is charged once, including reused common data.

Passive capture, identical across all Part 2/3 recording conditions, saves the already extracted overview/wrist RGB at every step without rendering or advancing physics. Evaluation-only numerical targets reproduce the native float32 local $+Z$ tilt calculation and per-object thresholds, plus negative target-to-goal distance. Targets never enter the live controller or dynamics input. Numerical uprightness is audited against native Boolean bindings. The final image remains saved but is omitted from model pairs without its missing subsequent two-finger proposal frame. A spatial distance readout is distinct from the native held-object goal; tilt is distinct from spill/drop safety.

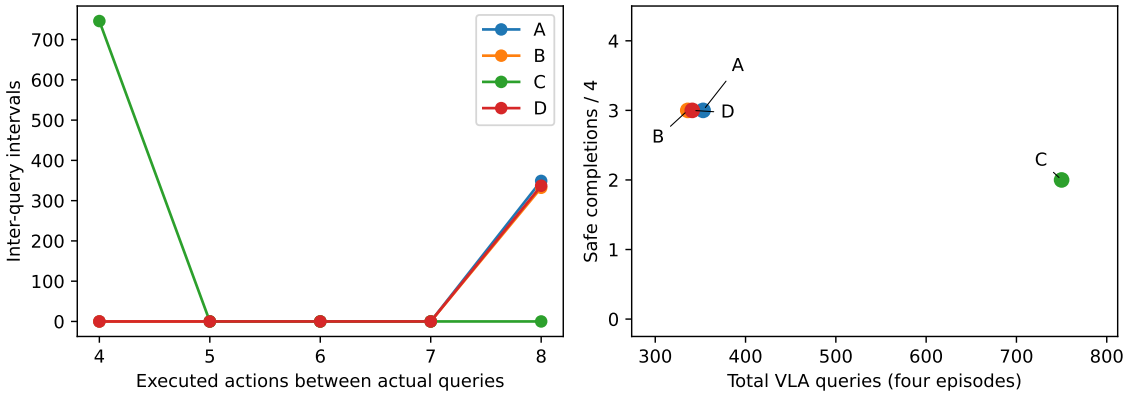


Figure 6: Observed inter-query intervals and query cost in Part 2. A/B/D always wait eight actions; C always waits four. Safe completions have denominator four.

Frozen predictive system. A pinned DINOv2 small encoder [6] produces 2×2 spatially pooled patch features from each camera, 3,072 coordinates in total. Fitting-only standardization and randomized PCA retain 32 visual coordinates. Seven measured arm joints and both finger positions remain separate. Five episode-bootstrap 128/128-unit residual MLPs predict the next standardized visual/robot state, conditioned on the current and previous state and previous/current actual 8-coordinate environment inputs. Each trains for 30 epochs with fixed settings. History shifts causally during rollouts of at most four actions. PCA is compression; the action-conditioned residual networks supply the dynamics.

Ridge heads estimate minimum upright margin and short-term progress. The audit compares persistence, fitted ridge, action-shuffled ridge and the same ensemble with mispaired action sequences, changing-state windows, and actual versus predicted future-state readouts. Raw pooled features, PCA and a same-size random projection diagnose representation/readout limitations. Correctly paired and shuffled sequences share the same trained ensemble in the action-dependence gate. Pre-fitting source refinements preserve their original receipts; numerical criteria, data groups, physical slots and model hyperparameters are unchanged.

Reliability and prospective search. M trusts ensemble disagreement; C uses an empirical conditional .8-error-quantile lookup by factual correction, with sparse bins falling back to the unconditional quantile and unsupported contexts abstaining. Both use nominal .5 calibration acceptance; actual acceptance and consequential errors must also be reported. These are empirical allocation heuristics, not calibrated safety probabilities. Planned counterfactual correction never becomes a factual residual without execution.

If the sealed forecast criteria permit, the 64-episode physical pilot evaluates a shared four-controller initial population on two cases (eight episodes once), three four-slot descendant populations on two fresh cases (24), two shortlisted controllers per method on two confirmation cases (12), and identity, one fixed Part 2 reference and three frozen finalists on four fresh blocks (20). Each method retains an actually re-evaluated elite, one unguided timing mutant and two cap mutants from a common pool of at most sixteen proposals. The known four-command prefix cannot rank later query timing, so timing genes stay in physical exploration. Abstention removes a model recommendation rather than a candidate's exploration eligibility. All deployed controllers share the same information and family; actual mean U determines selection and confirmation. No learned online action veto is installed. Three additional 32-action engineering attempts are available before eligible search, with no full-episode replacement. The deployed family has 160 genotypes: free caps $\{\varnothing, .12, .08, .04\}$ rad, grasp caps $\{\varnothing, .08, .04, .02\}$ rad, thresholds $\{0, .5, 1, 2, \varnothing\}$, and minimum intervals $\{4, 6\}$, with maximum eight and smoothing one fixed.

7.1 Part 3 forecast audit and scientific gate

The eight common recordings and all model workers complete normally. The forecast audit fails accuracy, with every support requirement satisfied; the physical 64-episode optimizer comparison is therefore unstarted by protocol. No Part 3 finalist, lineage, optimizer outcome or superiority result is asserted. This is a measured negative feasibility result, not a time-limited incomplete stage.

The corpus has sixteen fitting, four calibration and four audit episodes. Audit contains 552 four-action query-start windows, of which 263 exceed the fitting median displacement, 101 are near the upright boundary and 99 end with negative margin. Near-boundary events occur in four calibration and two audit episodes, with one audit episode crossing the boundary. There are 152 varying-progress windows. Windows are correlated within four episodes on one held-out task; they are not 552 independent safety trials.

On changing windows, one-action error is .2962 for the ensemble versus .2318 for persistence. Four-action errors are .4120 for persistence, .5179 for ridge, .5214 for action-shuffled fitted ridge, .6145 for the same ensemble with shuffled action prefixes and .5979 for correctly paired actions (Figure 7). The ensemble is 27.8% and 45.1% worse than persistence at one and four actions. Its 2.69% improvement over mispaired actions passes the action-dependence

Table 5: Frozen forecast criteria and observed values. All event-support criteria pass; seven accuracy/usefulness criteria fail. Degrees and millimetres are physical readout errors; latent errors use standardized coordinates.

Criterion	Measured	Required
One-step improvement	−27.8%	≥ 5%
Four-step improvement	−45.1%	≥ 5%
Shuffled-prefix improvement	2.69%	≥ 2%
Ensemble/ridge error	1.155	≤ 1.05
Actual-state margin MAE	16.17°	≤ 10°
Near-boundary forecast MAE	18.56°	≤ 10°
Progress MAE	79.01 mm	≤ 30 mm
Progress ranking, Spearman	.126	≥ .25
False reassurance	0/99	≤ .10
M/C actual acceptance	507/552	≥ .10
Sixteen-candidate screening	.810 s	≤ 30 s

check, but does not compensate for absolute/reference failures (Table 5). The actual-state head also fails before dynamics prediction is introduced.

All 818 calibration prefixes fall below the first .5 correction boundary, with maximum .17839. Sparse-bin fallback makes every conditional error quantile equal to .80913. C is then a positive affine transform of M; the same support conditions and transformed median thresholds produce exactly equal audit acceptance masks: 507/552 each, or 91.85%, versus 50% on calibration (Figure 8). Audit bins contain 551/1/0/0/0/0 prefixes. This calibration supplies no distinct correction-dependent allocation rule. It is retained rather than repaired using held-out outcomes. Zero false reassurance among 99 correlated negative-margin windows from one episode does not validate a safety guarantee.

A post hoc, frozen-weight diagnostic matches every head on the same 552 actual future observations. Margin MAE is 16.17° for PCA plus robot state, 11.03° for raw pooled features plus robot state, 12.82° for a random projection plus robot state, and 8.54° for visual-only PCA. Their progress errors are 80.57/52.94/104.53/41.13 mm. The visual-only contrast motivates testing cross-task covariate conditioning; it prevents diagnosing PCA information loss alone as the established cause. No head is replaced after audit. Fitting PCA retains 82.05% of visual variance, without certifying task/safety information.

The common recordings yield 3/8 safe completions, 4/8 violations and 5/8 unfinished episodes, with no unsafe completion. Calibration has 0/4 completions and three violations; audit has three safe completions and one violation. These data-acquisition outcomes are not an evolutionary comparison. Both native cameras, eight matched recording videos and complete event/episode picture strips are preserved.

8 PART 4: COUNTEREXAMPLE ACQUISITION PREMISE

Part 4 asks whether diverse physical hazards confidently missed by a small predictor can improve held-out prediction per acquisition cost. The planned comparison is stratified random sampling (R), uncertainty-guided mutation (U), concentrated missed-hazard evolution (E), and diversity-preserving evolution (D). Eight shared seeds, twelve new slots per method, eight confirmations and sixteen common audit episodes give an 80-episode ceiling, conditional on predictor, scene/reset and throughput feasibility. Policy and execution remain fixed; the predictor observes execution without changing actions.

The ingredients have precedents: quality-diversity scenario generation [2], uncertainty-based VLA annotation acquisition [5], and counterexample-guided joint agent/critic repair [1]. Their tasks and labels differ. No novelty or optimizer-superiority claim follows from implementing these ingredients.

8.1 Frozen causal target and direct predictor

The offline protocol is pushed at `e66c2f9` before the Part 4 label audit; the initial implementation follows at `9c9ccd9`. We restore and hash-check the exact 24 Part 2/3 feature records and corresponding traces from private dataset revision `f070abcb707b`. Its 3.34 GB raw archive passes SHA-256 verification. Pinned native object scopes and tilt thresholds agree with numerical margins in 37,362 proposition-observation checks. Robot measurements, proposals, actual environment inputs and query flags match their exact recorded trace entries.

The target is the first native uprightness violation of any protected object. At observation t, predict its occurrence in the next $H = 8$ actions while no violation has yet occurred. A useful warning must occur in $[\tau-8, \tau-4]$, where τ is first onset. At 20 Hz these specify a 0.4 s horizon and a 0.2 s deadline within the fixed-eight execution cadence. History ends at t; actual executed commands stop at $t-1$. We exclude already violated states, insufficient history and windows without eight real future observations. There is no terminal padding or unsuccessful-episode label substitution. One episode supplies at most one first-event warning denominator.

Fit-only PCA reduces the frozen two-camera DINOv2 features from 3,072 to 32 values. The 255-dimensional primary input combines the current and four preceding visual/robot observations, current proposed command, four previous actual commands, known controller settings and query age. Controller metadata conditions the historical A/B/C/D mixture. Privileged object states, monitor predicates, event time, task identity, scene genes and future observations/actions are excluded.

Five episode-bootstrap MLPs each have one 64-unit tanh layer and exactly 500 Adam updates of batch size 256. Bootstrap sampling is stratified by whole task. Per-trajectory window contributions are capped at 256, with retained positives, inverse-inclusion-weighted sampled negatives and equal episode mass before class balancing. Fixed L2 logistic models provide full-input and visual-only references. Platt probability calibration and each operating threshold use the common calibration task. Architecture, seeds, updates, targets and thresholds are not retuned using development outcomes. Encoder, dynamics and VLA weights stay fixed.

Fitting uses tasks 0039/0010, calibration 0046 and development 0006. All four fitting events occur on task 0039, failing the minimum two event-task criterion. Development has one event, failing

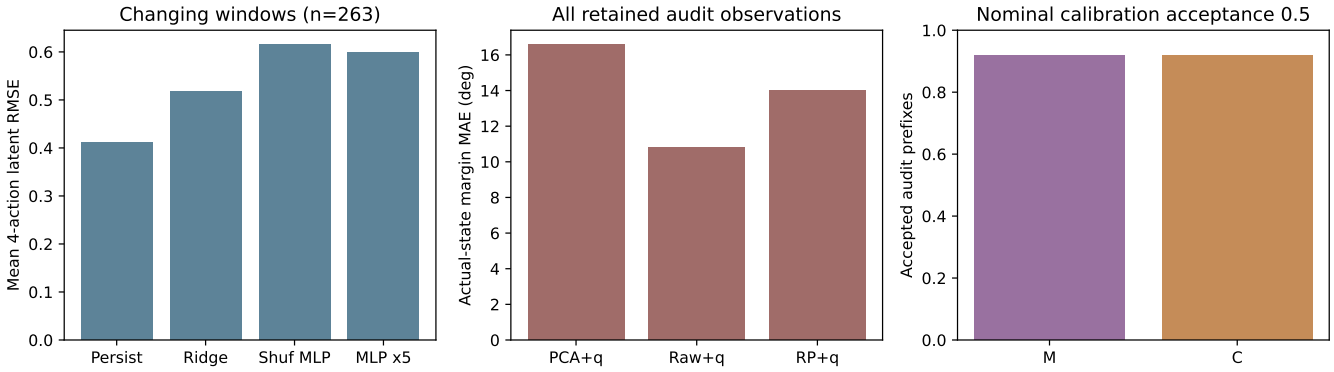


Figure 7: Part 3 forecast audit. Left: mean per-window standardized visual-latent RMSE on 263 changing windows. The shuffled-prefix control uses the same ensemble. Middle: actual-state readouts on every retained audit observation, with robot state included. Right: actual acceptance over all 552 audit prefixes. These quantities are not safety probabilities.

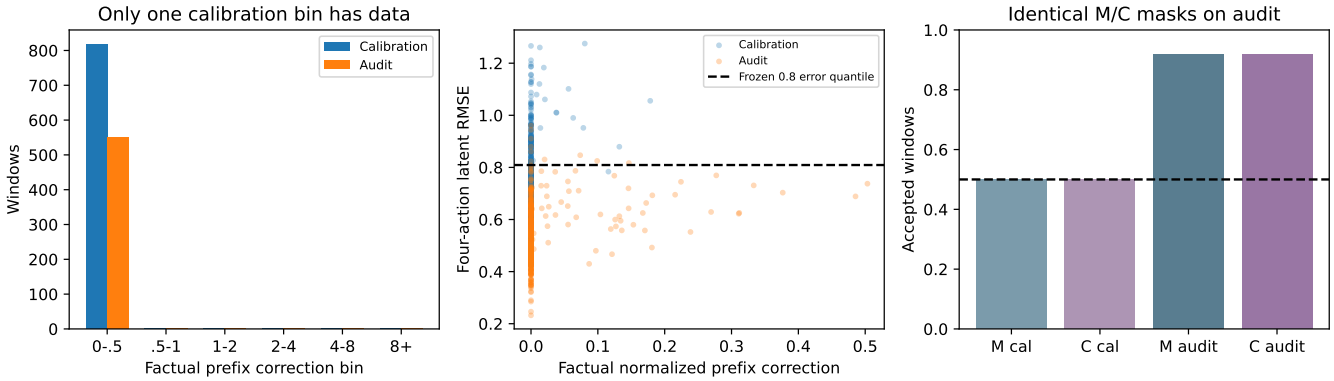


Figure 8: The frozen correction calibration collapses to one populated bin. Empty bins use the unconditional .8-error quantile. M and C have equal acceptance masks, and their nominal calibration acceptance does not persist on the audit task. No additional episode or changed threshold follows this result.

Table 6: Part 4 exposed historical development data. Positive windows are correlated within the first event; the Part 3 audit is not a fresh Part 4 audit.

Group	Tasks	Episodes	Events	Pos./windows
Fit	2	16	4	32/8,390
Calibration	1	4	3	24/2,584
Development	1	4	1	8/2,237

its minimum two episode criterion (Table 6). Missing support remains inconclusive. All task variants stay in one group, the fresh final audit is not selected or opened, and released VLA pretraining overlap remains unknown.

8.2 The operating point fails to generalize

The primary alarm threshold is .02995514. Its calibration false-alarm rate is 4.9801% when episodes receive equal weight; pooled counts are 110/2,560, or 4.2969%. Low risk means probability

Table 7: Part 4 development prediction at operating points frozen on calibration. False alarms are pooled negative windows. Every model warns on the one event; this alone does not establish a useful trade-off.

Predictor	AUROC	AP	False alarms
MLP ensemble	.8345	.0121	2,045/2,229
Full logistic	.8845	.0165	948/2,229
Visual logistic	.9735	.4258	1,575/2,229

at most .1. Low disagreement means uncalibrated member-probability standard deviation at most .00204003, the calibration negative-window 25th percentile. Calibration of the mean and the disagreement statistic are recorded separately.

Primary average precision (AP) exceeds the .003576 positive-window prevalence but fails the frozen .05 minimum. Its 91.75%

negative-window false-alarm rate exceeds the .15 maximum (Table 7). Every development episode has a false alarm, 4/4, as does every calibration episode, 4/4. Always-alarm also misses zero events but alarms on all 2,229 negative windows; always-safe misses 1/1 events without false alarms. The observed 0/1 missed timely warnings therefore cannot validate this operating point. The visual-only reference ranks better but still alarms on 70.66% of negative development windows; it is not substituted after the gate fails (Figure 9).

The primary misses 3/3 calibration events. One is a qualifying low-risk miss: `part3-record-b0-C`, bowl_69 onset 843, with deadline probability .0119821 and disagreement .00165808, giving descriptive quality .988018. The other two misses have greater disagreement. The sole development event, teacup_73 onset 307 in `part3-record-b3-B`, warns eight actions early, with deadline probability .113333 and disagreement .196481. It is not a confident miss; all 295 negative windows in that episode alarm. The calibration case neither proves reproducible acquisition nor supplies independent development support (Figure 10).

The frozen gate is `failed_discrimination`, with additional support and confident-miss failures. Shared-initial rescue requires discrimination to pass, so those eight episodes are ineligible. Physical scene/reset engineering, all 48 acquisitions, eight confirmations, repair and sixteen audit episodes remain unstarted. Empty stage receipts retain those distinctions; there is no generated scene, archive elite, repaired predictor, recurrence rate or held-out repair result. The CPU acquisition kernels and matched-update helper are implemented, with tests for bookkeeping and information isolation, but have no physical validation here.

8.3 Costs and preservation

Part 4 adds zero simulator attempts, controls, physics steps, VLA queries or encoder calls. The 24 reused donors retain their original 18,657 controls, 3,165 queries and 114,006 known post-subscription physics ticks, charged once. Their physical outcomes are 14 safe completions, one unsafe completion, two safe unfinished and seven violating unfinished episodes. Pre-subscription physics remains unknown; the uprightness prediction target does not replace the other native physical outcomes. Uprightness violations occur in 4/16 fitting, 3/4 calibration and 1/4 development episodes. Spill occurs in 3/4 calibration episodes; no donor has a native drop violation. Types can overlap. Task-specific proposition names are resolved from their pinned definitions and the resulting violation flag matches every audit.

New fitting uses 2,500 MLP updates and 640,000 sampled windows. Scoring evaluates 66,055 ensemble-member windows and 26,422 reference windows. One worker fails after training when reloading a partially reconstructed PCA estimator. Prediction resumes from identical hash-verified weights using the explicit saved affine transform; no second fit or model search occurs. The failed attempt and time are retained.

The existing replacement allocation is single-RTX-3090 instance 56676451, beginning 19 October 05:28:29.253 UTC, rather than the proposed 4090/5090. The physical cutoff is 12:28:29.253 UTC and hard stop 13:28:29.253 UTC. All preparation, implementation, reporting and idle time count from the original provider start, with an independent supervisor-managed deadline guard. This preparation base is not a validated ManiGuard stack, and no simulation-throughput or reset-repeatability claim is made.

Table 8: All physical attempts, including prior experience once. The total is the sum of known counters; one pre-application followup failure has unknown counters.

Stage	Attempts	Controls	Queries	Ticks
Part 1, all	52	35,158	3,401	215,420
Part 2, engineering	4	96	16	834
Part 2, comparison	16	11,190	1,780	68,516
Part 3, recording	8	7,467	1,385	45,490
Part 4, offline only	0	0	0	0
Total	80	53,911	6,582	330,260

Eight two-camera event strips and two 101-frame, 20 Hz videos use individually hash-checked historical RGB. They show warning start, deadline, native onset and post-onset observations without replaying physics. Own weights and evidence are separately private, revision-pinned and download/hash verified; full identifiers are in the Part 4 export and gallery linked from the paper README. Final preservation and provider state are recorded separately from scientific termination. This bounded result supplies no evolutionary acquisition or repair-efficacy claim.

9 CUMULATIVE ACCOUNTING AND PRESERVATION

Parts 2 and 3 reach their accurately reported terminal outcomes. The complete allocation contains eighty physical attempts: Part 1's 52 once, Part 2's twenty including engineering, and eight common Part 3 recordings (Table 8). The sixteen Part 2 fitting episodes are reused rather than recharged as new model interactions. The conditional 64-episode search and its three engineering checks remain unstarted. Known totals are 53,911 controls, 6,582 VLA queries and 330,260 post-subscription physics ticks. One pre-application followup failure has unknown native counters; all pre-subscription initialization physics remains unknown. These totals are recorded lower bounds rather than zeros for missing counters. An interim prose total of twenty Part 2 engineering queries is corrected to the audited $4 + 8 + 4 = 16$; raw records and outcomes are unchanged.

Scientific feature extraction encodes 37,314 images in 594 calls in 127.03 s. Training uses 11,158 transitions, 6,600 optimizer steps and 1,668,720 member examples, taking 16.40 s. Calibration, audit and candidate-screening use 380 member calls over 48,680 examples. Synthetic CPU engineering adds 720 member calls and 22,440 examples, 1,520 training examples and ten optimizer steps, plus a two-image encoder check; it is not empirical prediction evidence. The post hoc matched readouts add four matrix calls over 2,208 examples, without new data or dynamics/encoder/VLA queries. Torch extraction/training memory peaks are 539.7/80.7 MB; whole-GPU sampled memory and process overhead remain separate in the public accounting receipt.

The original 18 October 16:00:37 UTC clock is retained. The user explicitly extends the physical target to 19 October 03:00:37 UTC

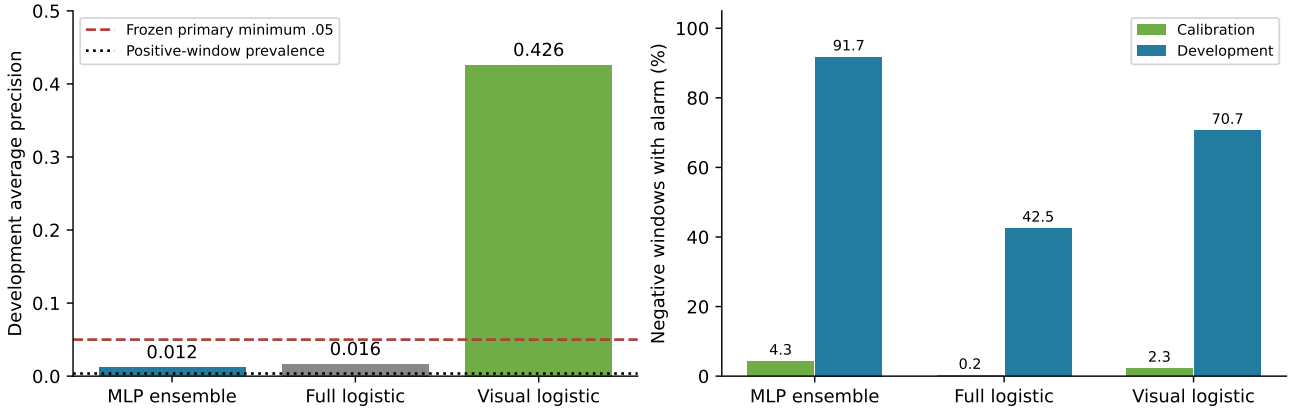


Figure 9: Part 4 discrimination and actual false alarms on exposed development data. The visual-only ablation ranks the one event better, while its frozen operating point still has excessive false alarms. Eight positive windows come from one event, not independent failures. Calibration columns show pooled rates; threshold selection weights episodes equally.

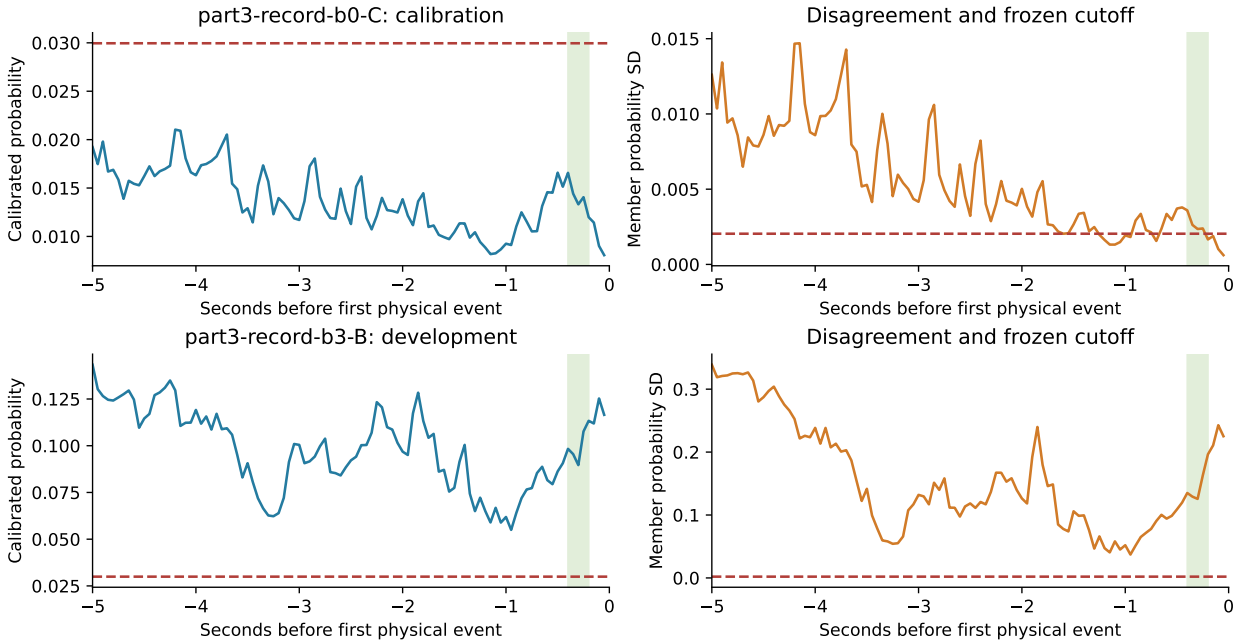


Figure 10: Historical calibration low-risk miss (top) and development event with excessive alarms (bottom). Green marks the eligible 0.4 to 0.2 s warning interval; red dashed lines show frozen probability and disagreement cutoffs. Scores end before the first event. No hypothetical action branch or new physics is used.

and hard end to 04:00:37 UTC. Setup, implementation, failures, reporting and idle rental all count; no study starts a new clock (Figure 11). Scientific workers and policy are stopped, with empty GPU compute processes independently verified at 5.538 allocated hours. Final rental time continues through preservation and the provider stop request; its actual/intended status receipt distinguishes an acknowledgement from a verified stop. The public cumulative ledger and closeout receipt carry the final cutoff.

New evidence contains 54 native camera clips, sixteen matched videos and 48 complete full-episode camera strips. Every violation has before/at/after images, while other episodes have start/middle/final images. The companion reports and complete gallery are linked from the paper README. The 3.34 GB raw archive, including RGB and features, is private and download/hash verified at dataset revision `a971f56192cd`. The own trained artifacts are separately private and pinned at model revision `e76969467146`; full revisions and file hashes are in the exports.

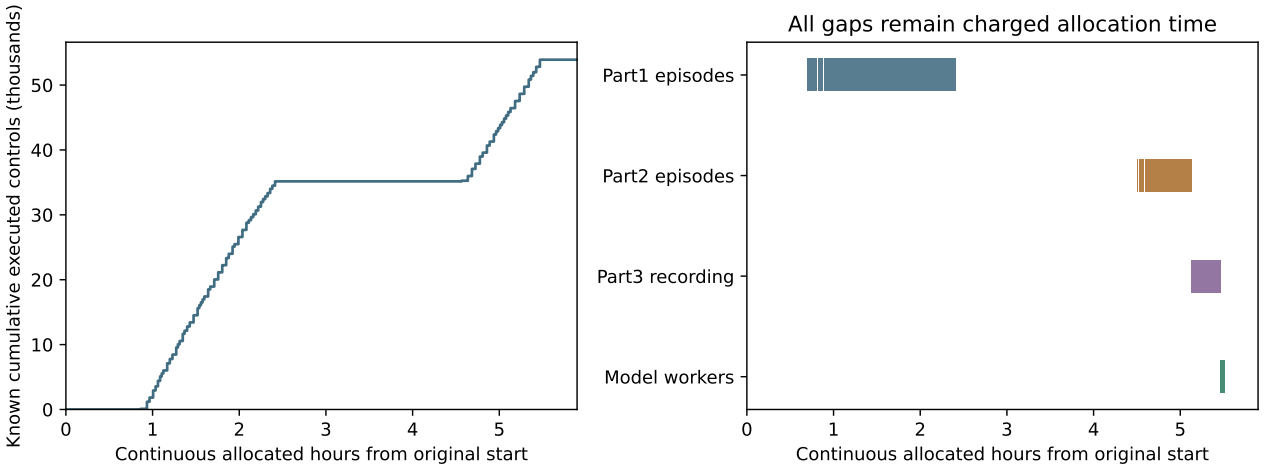


Figure 11: Known controls and episode/model process intervals against the original continuous allocation clock. Gaps include setup, implementation, reporting and idle rental and remain charged. Model replay adds no physical experience. The graph is an accounting snapshot, not a GPU-utilization estimate or rental-stop confirmation.

Assets, credentials and external checkpoints are excluded. Part 1's original episode artifacts, protocol, lineage and adverse findings remain preserved.

10 INTERPRETATION AND NEXT DECISION

The Part 1 training gain fails to transfer; Part 2 supplies no adaptive benefit because its trigger never changes timing; Part 3 fails factual prediction accuracy and correction-dependent calibration does not distinguish its proposed method from ordinary disagreement. These are separate findings. Neither a timeout nor any one case identifies a unique failure cause. The eight common data episodes and simple forecast references supply direction while avoiding an unsupported physical ranking experiment. No optimizer-superiority, calibrated-risk or safety-guarantee claim follows.

Part 4 retains a calibration low-risk miss but fails its separate development warning trade-off and event support. It therefore supplies no acquisition, diversity or repair comparison. The stronger visual-only ranking remains an information-ablation lead with excessive false alarms at its frozen operating point. Neither the absent development confident misses nor the unexecuted search establishes that counterexample acquisition cannot work.

A later study should establish saved-input repeatability and useful trigger activation on designated development data, validate prediction/readout generalization across whole tasks, and require supported correction strata that make the proposed trust rule distinct. The visual-only contrast motivates a readout information ablation. These are hypotheses for a separately frozen study, not fixes demonstrated here. Ordinary task-policy learning remains necessary for the broader contribution claim.

ACKNOWLEDGMENTS

AI coding assistance was used for implementation, analysis code, prose and typesetting. Experimental outcomes derive from the recorded simulator/monitor traces, and every experimental picture is extracted from recorded native video or lossless captured RGB. The paper adapts the repository's existing GECCO template without changing its archived experiment or results.

REFERENCES

[1] David Boetius and Stefan Leue. 2024. Counterexample-Guided Repair of Reinforcement Learning Systems Using Safety Critics. https://doi.org/10.48550/arXiv.2405.15430 arXiv:2405.15430
[2] Matthew Fontaine and Stefanos Nikolaidis. 2021. A Quality Diversity Approach to Automatically Generating Human-Robot Interaction Scenarios in Shared Autonomy. https://doi.org/10.48550/arXiv.2012.04283 arXiv:2012.04283
[3] Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. 2019. When to Trust Your Model: Model-Based Policy Optimization. https://doi.org/10.48550/arXiv.1906.08253 arXiv:1906.08253
[4] Bryan Lim, Luca Grillotti, Lorenzo Bernasconi, and Antoine Cully. 2021. Dynamics-Aware Quality-Diversity for Efficient Learning of Skill Repertoires. https://doi.org/10.48550/arXiv.2109.08522 arXiv:2109.08522
[5] Jie Ma, Zongxi Liu, and Yi Zhu. 2026. FailureSpot: Label-Efficient Timestamp-Level Failure Detection for Vision-Language-Action Models. https://doi.org/10.48550/arXiv.2609.04277 arXiv:2609.04277
[6] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, et al. 2023. DINOv2: Learning Robust Visual Features without Supervision. https://doi.org/10.48550/arXiv.2304.07193 arXiv:2304.07193
[7] Yiyan Peng, Philip Wang, Simon Sinong Zhan, Yiqi Lyu, Zhenyang Ni, Jixin Yan, Fiorelli Wong, Ruochen Jiao, Hang Yin, Xinyu Cao, Huajie Shao, Manling Li, Ruohan Zhang, and Qi Zhu. 2026. MANIGUARD: A Benchmark and Data Suite for Specification-Grounded Safety Evaluation and Improvement of Robotic Manipulation. https://doi.org/10.48550/arXiv.2608.17386 arXiv:2608.17386 [cs.RO]
[8] Physical Intelligence. 2026. openpi: Pinned policy implementation. https://github.com/Physical-Intelligence/openpi/tree/215abfb217dbac7d5f1273282331b9b1866c0479 Revision 215abfb217dbac7d5f1273282331b9b1866c0479; accessed 18 October 2026.
[9] Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. 2024. DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning. https://doi.org/10.48550/arXiv.2411.04983 arXiv:2411.04983